

At the eyepiece

What the Eye Still Knows. Thomas J. Watson Fellowship application, Aliaksandr Melnichenka, Berea College.

Everything on these two pages I made myself, with a phone held to an eyepiece or set on a wall. In Belarus the phone was a Redmi Note 7. The 2021 pictures were stacked in Sequator and adjusted in Lightroom. Page 1 is before Berea, page 2 is at Berea, and the people along the route follow on page 3.

Before Berea. Mazyr and the family's country houses, 2020 to 2021.



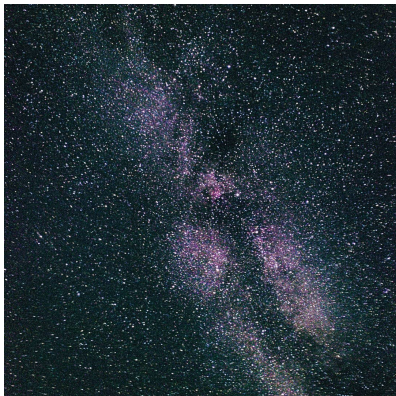
The Moon through my first telescope, the phone held to the eyepiece. Mazyr, August 2020.



The telescope itself, a small refractor at the open window of my room on the fifth floor. Mazyr, June 2020.



One raw frame of the Milky Way, 32 seconds at ISO 3200, before stacking. Our country house, 10 August 2021.



Twenty-one such frames stacked into one picture. My friends asked whether it was Photoshop. August 2021.



Star trails and a meteor over the lake near my grandfather's house, stacked phone exposures. August 2021.



Fifty exposures of 32 seconds, nearly half an hour of sky, over my grandparents' house. August 2021.

At Berea. The physics department's 8-inch telescope, an iPhone held to the eyepiece, 2024 to 2026.



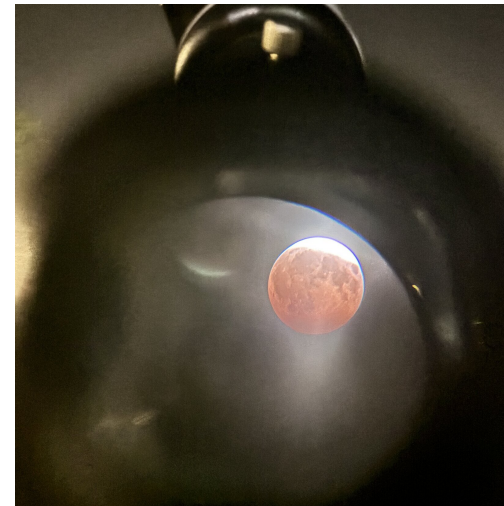
A visitor at the 8-inch on the observing deck of the science building during International Observe the Moon Night, which I helped set up. 14 September 2024.



The Moon through the 8-inch that night, the phone held to the eyepiece. 14 September 2024.



The Orion Nebula through the 8-inch on the same deck. 24 February 2025.



The partial lunar eclipse in the eyepiece during the Physics Club's public night on the alumni field. 27 August 2026.

Hosts and correspondents

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I wrote to the observatories, societies and individual observers along the route of this year. The table lists the replies the year rests on, in the order I will meet the people, with the substance of what each one wrote. Words in quotation marks are theirs. The ten people in the application's project contacts are marked with an asterisk.

Where and when	Who	What they wrote
Egypt, 22 July to about 20 August 2027. The Luxor eclipse of 2 August.		
Kottamia and Luxor, July and August 2027	Dr. Bassem Nabawy* President, National Research Institute of Astronomy and Geophysics (NRIAG), Egypt's national institute, which runs the 1.9-meter telescope at Kottamia, the largest in North Africa	"You are welcome to Katamiya Observatory." One night at Kottamia between 22 and 28 July 2027 with hosting, at the institute's fee, and a place at its eclipse camps at Luxor, Qena or Marsa Alam once the camp fees are set, which he is still estimating "due to the very high rate of booking in Luxor."
Luxor, 2 August 2027	Dr. Awni Al-Khasawneh Secretary General, Arab Union for Astronomy and Space Sciences, the union of the Arab countries' astronomical societies	Confirmed that the Union and its member societies will observe the eclipse at Luxor, that he will personally supervise them, and that I may coordinate my participation directly with him. He called it "your inspiring fellowship project" and wrote, "We are pleased to support your project."
Luxor, eclipse week	Prof. Zainab Awad Cairo University, co-chair of the university's eclipse conference in Luxor	"You're most welcome to join the conference and attend the observations session." I may keep my own logbook during the session. She also asked how I had heard of the conference before its public announcement.
Luxor, the evening of 2 August	Dr. Steve Bell HM Nautical Almanac Office, UK Hydrographic Office	The Crescent MoonWatch project continues, "positive and negative sightings are always useful," and he listed what a report should carry, from the weather and the orientation of the horns to the observer's experience, where "First Time" is one of the boxes.
Luxor, the evening of 2 August	Prof. Bradley Schaefer Louisiana State University	"I am positively impressed with your idea of providing an outside observer for the Egyptian committee." He spent "a man-year of work" on the model of crescent visibility, explained why the false-alarm rate is the point of standing beside such a committee, and told me to expect little notice of my report. "Do what you think best, and act wisely."

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Chile, late August to October 2027.		
Elqui Valley, August 2027	Gabriela López* Head, Office for the Protection of the Quality of the Sky of Northern Chile (OPCC), which keeps the sky dark over Chile's great observatories, with Juan Pablo Uchima	"You can count on our support to accompany us during night sky brightness measurements." The round is made near the new Moon at a professional observatory, always, and the office measures every month whatever the winter does. Uchima added, "I assume you are used to cold weather, so that will probably not be the main issue."
La Serena, September 2027	Javier Sayago NOIRLab, La Serena, lighting audits under the region's dark-sky law	"No hay problema." An outing or a demonstration of how the lighting standard is checked, to be fixed nearer the date. On my Spanish, "we speak English too."
Vicuña, August to September 2027	Juan Daniel Durán Rojas* Office of the Mamalluca municipal observatory, Chile's best-known public observatory, open since 1998	"Su propuesta nos parece muy interesante." Willing in principle to have me alongside the observatory's guides and astronomers, the details to be coordinated at the start of 2027. "Esperamos poder recibirlo en Mamalluca," we hope to receive you.
Las Campanas, one night	Dr. Juna Kollmeier* Director, Sloan Digital Sky Survey, whose three-dimensional maps of the universe are the most detailed ever made	"Yes, this can be arranged." One night beside the survey's observers at Las Campanas. She added that my adviser Alex Lazarian once promised her he could read the turbulent field from the survey's Local Volume Mapper data alone, "maybe you will prove that he was telling the truth!" She applied for a Watson herself as a Caltech undergraduate with a similar plan, did not get it, went to Germany on a Fulbright instead, and offered the visit whatever the outcome.
Las Campanas, August to September 2027	Carol Rojas Communications and outreach, Las Campanas Observatory, Carnegie Science	Answered with one question, the question of this whole year. "Would you rather talk to someone who knows the night sky by heart, rather than someone who works at an observatory and observes through a telescope?"
Cerro Pachón, pending	Kristen Metzger Vera C. Rubin Observatory, the new 8.4-meter telescope that will photograph the whole southern sky every few nights for ten years	Is asking the head of the observing team whether a night beside them on the mountain is possible now that operations have begun. The next days were holidays in Chile, "but I wanted you to know I'm looking into it."

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La Silla, September 2027	Dr. Petr Kabáth Astronomical Institute of the Czech Academy of Sciences, the PLATOSpec spectrograph at the European Southern Observatory's La Silla	A day visit "should be possible" if his team is on the mountain, to be settled in March 2027. The telescope is run remotely from Europe or Santiago, and "there is no eyepiece for observing by eye during the night."
Valparaíso, 13 to 20 October 2027	Domingo González President, Sociedad Astronómica de Valparaíso y Viña del Mar (SAVAL), an amateur society founded in 1956	"It will be a pleasure to receive you, share experiences and learn more about your outstanding career." The society celebrates its seventieth anniversary in Valparaíso this October, "and we would love to have you."
Argentina, November to December 2027.		
La Plata and Buenos Aires, November 2027	Dr. Sixto Giménez Benítez* Archaeoastronomer, La Plata, suggested by César González-García of the European Society for Astronomy in Culture	"Claro que puedo ayudarte," written from a congress in Rio de Janeiro. A La Plata visit in November 2027 should be no problem, help in Buenos Aires is easy, an afternoon on measuring a horizon marker's orientation by eye.
Buenos Aires, November 2027	Amigos de la Astronomía Founded in 1929, among the oldest amateur societies in Latin America. The association's secretary	Student membership with proof of enrollment. Members help with the association's activities, and with the guided visits once they know its history and its ways as well as the sky. "Recordá solicitar la baja cuando te vayas," remember to cancel the membership when you leave.
Barreal, 29 November to 8 December 2027	Dr. Antonio Damián Alejo* Félix Aguilar Observatory, National University of San Juan, whose Cesco station stands in El Leoncito National Park, created to protect the observatories' sky	Is arranging visual observing nights for me at the observatory's Cesco station at Barreal, 2,350 meters up in the Andes. "Barreal is a very special place for astronomy, and I think your project fits very nicely with the astronomical tradition of the area."
Peru, January to February 2028.		
Sacred Valley, 9 to 23 February 2028	Fernando Sánchez Montero* Sembrando Cielos, Calca	"In principle I would be happy to have you join us for a night." Help alongside his team and the Andean dark constellations by eye, Yacana the llama among them. "And if the weather does not allow us to observe the sky, you can still count on our support," an indoor session at his visitor space. Permission to name him given.

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Where and when	Who	What they wrote
Australia, March to May 2028, and the eclipse of 22 July 2028.		
New South Wales, autumn 2028	Prof. Ray Norris* CSIRO Space and Astronomy, and Western Sydney University. He leads the Evolutionary Map of the Universe survey on the ASKAP radio telescope and is a leading researcher of Aboriginal astronomy	Will put me in touch with a Gamilaraay elder once the elder gives permission. On the Emu in the Sky, “most people interested in this area will already be familiar with the shape of the emu in the sky, just from looking at the sky, but it may be from conversations with a knowledge holder that they learn about the subtleties of how it varies through the year.”
Dubbo, late April and 22 July 2028	Peter Starr Dubbo Observatory, New South Wales	“Yes we are planning to run a session for this and have some telescopes set up for it.” He offered to put me on the list for eclipse day, inland from Sydney’s winter cloud, and the observatory shows the sky through telescopes every night.
South Australia, if I detour	Andrew Cool River Murray International Dark Sky Reserve	Has measured the reserve’s sky brightness since 2018. “At our latitude, Sagittarius passes overhead,” and the new observatory at Swan Reach has a telescope “big enough to allow colour to be perceived in the Trifid nebula through the eyepiece.”
Sydney, 22 July 2028	Meredith Hall* Manager, Inspiring Australia (NSW), the national science engagement program, University of Wollongong	“Wonderful to hear from you and the Fellowship sounds like an exciting opportunity. We would be glad to assist with finding you a volunteer opportunity to be a part of the eclipse celebrations.”
Sydney, 22 July 2028	Dave Reneke Astronomy writer and broadcaster, New South Wales	The one rule he gives a class before a solar eclipse, “Glasses on, keep them on, follow the supervisor’s instructions,” how to run a pinhole projector for a group, and the calm radio announcement he would make as totality approaches Sydney.
New Zealand, May to mid July 2028.		
Kaikōura, 8 to 21 May 2028	Lisa Bond* Kaikōura Dark Sky Trust, keepers of an International Dark Sky Sanctuary certified in 2024	“You would certainly be very welcome to visit us.” A round of the sanctuary’s sky-quality readings with the team, and any community stargazing evening or school session in my two weeks. “Being in the Southern Hemisphere will also give you a wonderful opportunity to experience objects in our night sky that you won’t see from your part of the world.” Permission to name the Trust given.
Christchurch, May 2028	Prof. John Hearnshaw University of Canterbury	In principle happy to meet in Christchurch in May 2028, on how visual magnitude estimates were checked, and he asked whether email or Zoom would not serve the question better. I still want the hour in person.

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Where and when	Who	What they wrote
Wairarapa and Wellington, May to June 2028	Dr. Haritina Mogoşanu Star Safari observatory and Spaceward Bound New Zealand	“Fantastic to hear from you and great credentials.” Introductions to the Victoria University physics team and to teachers across the country, her observatory “in the best darkest place in New Zealand,” the national astronomy olympiad team she runs, and an offer of a letter. “Yes you can say that.”
Wellington, May to July 2028	Adam Rosner Wellington Astronomical Society	The society’s meetings are public. Star Field nights at its dark-sky site on 20 or 27 May, 17 or 24 June and 22 July 2028, with a ride shared over the mountains. Camping there in June means nights below freezing, “Translation: It gets cold at night,” so he pointed me to the town of Martinborough.
Wellington, May 2028	Briar Visitor Experience Lead, Space Place, Wellington	“We’d be happy to have you along to observe and support with operating the telescopes on one of those nights.” The staff always find the target for the visitor, and “sometimes young children find it hard to see through the eyepiece.”
Naseby, June to July 2028	Paul Bishop Night-sky tours, Naseby, Central Otago	Almost every evening of June and July 2028 free, the only exception being the eclipse itself, and a 12-inch telescope that can be driven by hand. “That sounds perfect!”
Lauder, June 2028	Richard Querel Lauder atmospheric research station, Earth Sciences New Zealand	“Yes, we host visitors and can give a tour.” A chance to watch an ozone balloon launch if the weather is calm, usually on a Monday or Tuesday. The balloons are two meters across when filled, followed by eye until they are a dot about a kilometer up, and burst near 34 kilometers. “Your BUSEL-1 project looks interesting.”
Lower South Island, June 2028	Ian Cooper Aurora observer by eye since 1978, Royal Astronomical Society of New Zealand	Sent the society’s visual report form, which he updated in 2024, and how to write a sighting down. “The eye could see fine detail that was often blurred on film.” Auroras that only a camera picks up “we call subvisual,” and he uses his own camera just to confirm a glow he suspects.
Invercargill, June 2028	Elizabeth King President, Southland Astronomical Society	Welcome at the society’s monthly meeting and its public viewing night. She has chased the aurora for fifteen years, green on strong nights and red and pink on the best. “What I see with the naked eye others may not.”
Dunedin, 26 June to 15 July 2028	Jessica Hatfield and Georgette Watkins Tūhura Otago Museum	Volunteer opportunities in and around the planetarium. “Happy for you to mention we have opportunities available for you.”
Dunedin, 14 July 2028	Michelle Taiaroa University of Otago, Kāi Tahu	The Matariki dawn ceremony on the museum lawn, and how to attend it respectfully. “During the ceremony you will start to hear people call out names. They are the names of their loved ones that have passed away since the last Matariki.” Breakfast in the museum afterwards.
Dunedin, July 2028	Prof. Jonathan Squire Department of Physics, University of Otago	No public astronomy program there, but the department could host a visit. He knows Jim Fuller, my Caltech mentor, from his own Caltech days, and sent me to the Dunedin Astronomical Society, “very friendly and enthusiastic.”

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Where and when	Who	What they wrote
Advisers on the eye, for every leg.		
Anywhere	Stephen O'Meara Discovered the spokes in Saturn's rings by eye	How an eye trained to a tenth of a magnitude on variable stars found the spokes through a 9-inch refractor at Harvard. He tried for a Moon a few hours old in Mallorca this year and lost it in the smoke of mainland fires. "But you can always try."
Anywhere	Gary Poyner British Astronomical Association, 370,000 visual estimates	Why he stayed at the eyepiece when cameras arrived. "There is something very special in seeing a rare outburst of a dwarf nova, sometimes for the very first time." On a clear night he covers more than 250 stars by eye.
Anywhere	Joanna DiStefano American Association of Variable Star Observers, whose archive of visual estimates began in 1911	Student associate membership and the mentor program for learning visual estimates. Her mother went to Berea, and she herself once had dinner at Boone Tavern with Loyal Jones of the Appalachian Center.
Anywhere	Dr. Mitsuru Sôma Vice President for Grazing Occultation Services, International Occultation Timing Association	"Observations do not need a team. A beginner also can make a graze report." Reports accepted from anywhere in the world.
Anywhere	Prof. Trevor Lamb Australian National University, Fellow of the Royal Society, the physiology of dark adaptation	How long the rods take to recover after a glance at a screen, "possibly 10 mins," and why. "Do NOT use a phone/laptop if you want your best scotopic threshold." A deep-red filter if I must.
Anywhere	Ian Ridpath Author of Star Tales, editor of the Oxford Dictionary of Astronomy	How navigators found the south celestial pole by eye, down the long axis of the Southern Cross to where it meets the perpendicular from Alpha and Beta Centauri. "Amateur astronomers still use the same technique today."
Anywhere	Prof. Kevin Krisciunas Texas A&M University	The faintest star he has seen with the unaided eye, magnitude 6.3 from the Pyrenees in 2019, a candle flame at about 2.6 kilometers. "Maybe its color helped me detect it."
Anywhere	Prof. Matthew Bailes Swinburne University, co-discoverer of the first fast radio burst, Shaw Prize 2023	How the first burst was found at Parkes, a threshold in software and then a person looking at a plot. Today's searches run about a billion trials a second. "You can't keep the raw data."
Anywhere	Anthony Tekatch Unihedron, maker of the Sky Quality Meter	How to take a reading with the center of the Galaxy overhead. Record every variable, take many readings, and trust the zenith over the horizon.
Anywhere	Alina Sabyr 2019 Watson Fellow	"Your project sounds really fascinating." A Zoom call on a year that starts with an eclipse, once her busy weeks pass.